Universidad Distrital Francisco José de Caldas
School of Engineering

Data Analysis Programming
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Flood Monitoring and Risk (Texas)

Workshop No. 1: Hazard Definition, Source Validation, and Data Contract

Project Definition Report — Team 2

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Repository: `https://github.com/natcastill/flood-monitoring-team2`

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1 Hazard Definition

1.1 Phenomenon and why it matters

A *flood* is an overflow of water onto normally dry land; it can last days or weeks. A *flash flood* is a flood caused by heavy or excessive rainfall in a short period, generally less than six hours (National Weather Service, 2026a). Texas combines the conditions for both: intense convective and tropical rainfall, thin soils and steep terrain in the Hill Country, and large urban areas with extensive impervious surfaces (Houston, Dallas–Fort Worth, San Antonio, Austin). Flooding there causes recurrent loss of life (often in vehicles driven into flooded roads) and large property damage, so knowing quickly *where heavy rain is falling now*, *how unusual it is*, and *whether official warnings are active* is valuable for emergency-management decisions.

1.2 How it is measured and severity

Flooding has no single physical magnitude comparable to an earthquake's. The system therefore uses three complementary measures, each with explicit units:

- **Rainfall accumulation** (mm) over 1 h, 24 h and 7 days, from station observations (NWS) and daily totals (GHCN-Daily). This is the physical driver and the main input of the indicator.
- **Official alert level** (ordinal): Advisory/Statement < Watch < Warning, from NWS alerts (flood and flash-flood products).
- **Impact** of past events (deaths, injuries, property damage in USD) from the Storm Events Database, used to describe historical severity and to validate the indicator.

These are mapped to a common ordinal `severity_level` 0–3 while keeping the native value and unit (Section 5).

1.3 Geographic scope and time window

- **Scope: State of Texas (USA)**, bounding box 25.8–36.5°N, 106.65–93.5°W. Our required sources (Storm Events, NWS) are U.S. national services with no data for Colombia, so a U.S. scope is mandatory for Team 2. Texas has one of the highest counts of flood and flash-flood records in the country, which gives enough events for profiling, overlap analysis and indicator validation, while staying a single state (one NWS `area=TX` filter, one state FIPS code `48`).
- **Observation network:** five first-order stations that exist in both GHCN-Daily and the NWS API: Houston IAH (USW00012960/KIAH), Austin-Bergstrom (USW00013904/KAUS), San Antonio (USW00012921/KSAT), Dallas–Fort Worth (USW00003927/KDFW) and Corpus Christi (USW00012924/KCRP). The pairing is what allows comparing current and historical rainfall at the same place.
- **Historical window: 2010–present.** It provides 15+ years for a stable rainfall distribution and flood frequency per county, while keeping the bulk load small (17 yearly Storm Events files, 194.9 MB compressed; Section 2.2).

All four sources cover this scope: Storm Events and NWS alerts are filtered by state, and the selected stations are in Texas.

1.4 Definition of "current situation"

For this hazard, the current situation is: (1) rainfall in the **last 24 hours** and **last 7 days** at each station; (2) the set of **active NWS flood alerts** for Texas at query time; and (3) the comparison of (1) with the historical distribution for the same station and month. Storm Events is not "current": new months appear with several months of delay, so it is used only for history and validation.

2 Historical Sources

Table 1: Historical sources.

	H1: NOAA Storm Events	**H2: GHCN-Daily (NCEI ADS)**
Owner	NOAA NCEI	NOAA NCEI
URL	`https://www.ncei.noaa.gov/pub/data/swdi/stormevents/csvfiles/`	`https://www.ncei.noaa.gov/access/services/data/v1`
Access method	Bulk HTTPS download, one gzipped CSV per year (`details`, `locations`, `fatalities`)	REST query, `dataset=daily-summaries`, `stations`, `startDate`, `endDate`, `dataTypes=PRCP`, `units=metric`
Authentication	None	None
Rate limits	None published; files cached locally	None published for this use; requests spaced and cached
Format	CSV (gzip), 51 columns	JSON (or CSV)
Key fields	EVENT_ID, EPISODE_ID, EVENT_TYPE, BEGIN_DATE_TIME, CZ_TIMEZONE, CZ_TYPE/FIPS/NAME, BEGIN_LAT/LON, DAMAGE_PROPERTY, DEATHS_*, FLOOD_CAUSE	DATE, STATION, PRCP, PRCP_ATTRIBUTES (MFLAG, QFLAG, SFLAG, OBSTIME), NAME, LATITUDE, LONGITUDE, ELEVATION
Time coverage	1950–present (flood types from 1996)	Station-dependent, decades; used from 2010
Update policy	Files re-compiled when NOAA revises them (suffix `_cYYYYMMDD` changes); recent months with several months of delay	Daily; recent values can be added or corrected
License / attribution	U.S. Government work, public domain; cite NOAA National Centers for Environmental Information (2026b)	Public domain; cite Menne et al. (2012) and NOAA National Centers for Environmental Information (2026a)
Filter used	STATE = TEXAS; EVENT_TYPE in {Flood, Flash Flood, Heavy Rain}	5 Texas stations listed above

2.1 Evidence: real responses

H1 — Storm Events (directory listing request and first rows of the 2025 Texas sample, 724 flood-related rows; raw files in `workshop_1/samples/storm_events/`):

```
GET https://www.ncei.noaa.gov/pub/data/swdi/stormevents/csvfiles/
HTTP 200 retrieved 2026-11-06T02:20:01.339680+00:00
Content-Type: text/html; charset=UTF-8
```

```
# texas_floods_2025.csv (selected columns)
EVENT_ID,EVENT_TYPE,BEGIN_DATE_TIME,CZ_TIMEZONE,CZ_TYPE,CZ_NAME,BEGIN_LAT,BEGIN_LON,DAMAGE_PROPERTY
1256799,Flood,21-APR-25 10:30:00,CST-6,C,WILLACY,26.5513,-97.424,0.00K
1286074,Flash Flood,08-JUL-25 18:55:00,CST-6,C,DALLAS,32.82,-96.92,0.00K
1233050,Flash Flood,29-JAN-25 23:15:00,CST-6,C,DALLAS,32.81,-97.03,10.00K
1233052,Flash Flood,29-JAN-25 23:58:00,CST-6,C,ELLIS,32.49,-96.81,2.00K
```

H2 — GHCN-Daily (first records of one station; 1825 daily records in total; raw files in `workshop_1/samples/ghcnd/`):

```
GET https://www.ncei.noaa.gov/access/services/data/v1?dataset=daily-summaries&dataTypes=PRCP&units=metric&
    includeStationLocation=1&includeAttributes=True&stations=USW00003927&startDate=2025-01-01&endDate
    =2025-12-31&format=json&includeStationName=true
HTTP 200 retrieved 2026-11-06T02:20:12.762158+00:00
Content-Type: application/json;charset=utf-8

{"DATE": "2025-01-01", "STATION": "USW00003927", "LONGITUDE": "-97.02196", "PRCP_ATTRIBUTES": ",,W", "...
{"DATE": "2025-01-02", "STATION": "USW00003927", "LONGITUDE": "-97.02196", "PRCP_ATTRIBUTES": ",,1", "...
{"DATE": "2025-01-03", "STATION": "USW00003927", "LONGITUDE": "-97.02196", "PRCP_ATTRIBUTES": ",,W", "...
{"DATE": "2025-01-04", "STATION": "USW00003927", "LONGITUDE": "-97.02196", "PRCP_ATTRIBUTES": ",,W", "...
```

2.2 Volume estimate and bulk-load partitioning

- **Storm Events:** 17 yearly `details` files for 2010–2026 add up to 194.9 MB compressed (measured with HTTP `Content-Length`, `volume_estimate.csv`). The 2025 file contains 724 Texas flood-related rows, so the window holds roughly 12,308 rows after filtering. **Partition: one year per unit**, which also matches how NOAA re-compiles the data.
- **GHCN-Daily:** 5 stations $\times$ 365 days $\times$ 17 years $\approx$ 31,000 rows (a few MB of JSON). **Partition: station $\times$ year** (85 requests), small enough to retry individually.

3 Incremental Sources

Table 2: Incremental sources (both from the NWS API, National Weather Service, 2026b).

	I1: NWS station observations	**I2: NWS alerts**
URL	`https://api.weather.gov/stations/{id}/observations`	`https://api.weather.gov/alerts/active?area=TX` (and `/alerts?area=TX` for recent history)
Authentication	No key; mandatory `User-Agent` with application and contact (stored in `.env`)	Same
Format	GeoJSON; values as {`value`, `unitCode`, `qualityControl`}	GeoJSON features with CAP properties
Update frequency	Every 5 minutes at these airports (5-min ASOS reports), plus 15-min and hourly METAR reports; about 7 days retained	Continuous; issued, updated and cancelled at any time
Change detection	`timestamp` per station; `start` parameter; HTTP cache headers	Stable `id` per message; `sent`; `messageType` Alert/Update/Cancel and `references`
License	Public domain	Public domain

3.1 Evidence: real responses

I1 — observations (latest poll 20261106T032630Z; 2500 observations across 5 stations):

```
GET https://api.weather.gov/stations/KAUS/observations?limit=500
HTTP 200 retrieved 2026-11-06T03:26:32.494521+00:00
Content-Type: application/geo+json
Content-Length: 42407
Expires: Fri, 06 Nov 2026 03:31:32 GMT
Cache-Control: public, max-age=300, s-maxage=120

{
 "geometry": {
  "type": "Point",
  "coordinates": [
   -97.67,
   30.19
  ]
 },
 "properties": {
  "@id": "https://api.weather.gov/stations/KAUS/observations/2026-11-06T02:55:00+00:00",
  "stationId": "KAUS",
  "timestamp": "2026-11-06T02:55:00+00:00",
  "textDescription": "Clear",
  "temperature": {
   "unitCode": "wmoUnit:degC",
   "value": 18,
   "qualityControl": "V"
  },
  "precipitationLastHour": null
 }
}
```

I2 — active alerts for Texas (1 active alerts at poll time; an empty collection is a valid answer):

```
GET https://api.weather.gov/alerts/active?area=TX
HTTP 200 retrieved 2026-11-06T03:26:36.545360+00:00
Content-Type: application/geo+json
Content-Length: 1566
ETag: W/"7f9d71f0e5b22344160c7f90156d4d9c:dtagent10337260504112723F6xZ:dtagent10337260504112723F6xZ"
Expires: Fri, 06 Nov 2026 03:26:41 GMT
Cache-Control: public, max-age=5, s-maxage=5

{
 "geometry": "...",
 "properties": {
  "id": "urn:oid:2.49.0.1.840.0.44543a7ad9324c74ecf2274302ec477fdf42445c.001.1",
  "event": "Flood Advisory",
  "messageType": "Alert",
  "status": "Actual",
  "severity": "Minor",
  "sent": "2026-11-05T20:40:00-06:00",
  "onset": "2026-11-05T20:40:00-06:00",
  "expires": "2026-11-05T22:45:00-06:00",
  "areaDesc": "Nueces, TX",
  "references": []
 }
}
```

3.2 Empirical update behavior (two polls)

The incremental sources were polled at 20261106T022015Z and 20261106T032630Z (UTC), at least one hour apart, and compared with `scripts/compare_polls.py`:

```
# Incremental poll comparison

- Poll 1: ‘poll_20261106T022015Z‘
- Poll 2: ‘poll_20261106T032630Z‘

## NWS alerts (active + recent, Texas)
- Poll 1: 226 distinct alert ids; poll 2: 226
- New ids in poll 2: 1; ids no longer returned: 1
- Same id, different content: 0
- New alerts that reference earlier alerts (Update/Cancel chain): 0
- messageType in poll 2: Alert=115, Cancel=31, Update=80


## NWS station observations
| station | obs poll 1 | obs poll 2 | newest ts poll 1 | newest ts poll 2 | new ts | overlapping ts with ch...
|---|---|---|---|---|---|---|
| KAUS | 500 | 500 | 2026-11-06T01:15:00+00:00 | 2026-11-06T02:55:00+00:00 | 7 | 0 |
| KCRP | 500 | 500 | 2026-11-06T01:24:00+00:00 | 2026-11-06T02:55:00+00:00 | 8 | 0 |
| KDFW | 500 | 500 | 2026-11-06T01:15:00+00:00 | 2026-11-06T02:55:00+00:00 | 8 | 0 |
| KIAH | 500 | 500 | 2026-11-06T01:15:00+00:00 | 2026-11-06T02:55:00+00:00 | 7 | 0 |
| KSAT | 500 | 500 | 2026-11-06T01:15:00+00:00 | 2026-11-06T02:55:00+00:00 | 7 | 0 |

## HTTP headers
| endpoint | header | poll 1 | poll 2 | changed |
|---|---|---|---|---|
| alerts/active | ETag | W/"d190d23230a0aa478d1317904aebb117:dtagent10337260504112723F6xZ:dtagent1033726050...
| alerts/active | Last-Modified | - | - | no |
| alerts/active | Expires | Fri, 06 Nov 2026 02:20:24 GMT | Fri, 06 Nov 2026 03:26:41 GMT | yes |
| alerts/active | Cache-Control | public, max-age=5, s-maxage=5 | public, max-age=5, s-maxage=5 | no |
| alerts/active | X-Request-Id | - | - | no |
| alerts/active | Date | Fri, 06 Nov 2026 02:20:19 GMT | Fri, 06 Nov 2026 03:26:36 GMT | yes |
| observations/KIAH | ETag | - | - | no |
| observations/KIAH | Last-Modified | - | - | no |
| observations/KIAH | Expires | Fri, 06 Nov 2026 02:25:15 GMT | Fri, 06 Nov 2026 03:31:31 GMT | yes |
| observations/KIAH | Cache-Control | public, max-age=300, s-maxage=120 | public, max-age=300, s-maxage=120... |
| observations/KIAH | X-Request-Id | - | - | no |
| observations/KIAH | Date | Fri, 06 Nov 2026 02:20:15 GMT | Fri, 06 Nov 2026 03:26:31 GMT | yes |
```

What changed between the polls (about 66 minutes apart).

- **Observations:** each station returned its latest 500 reports both times; 7–8 new timestamps per station appeared and none of the overlapping observations changed. The newest observation was 30–65 minutes old at query time, so data are published with a delay. The responses carry `Cache-Control: max-age=300` and `Expires`, but no `ETag` or `Last-Modified`: new records must be detected by `timestamp`, not by HTTP headers.
- **Alerts:** active alerts for Texas went from 0 to 1 (a new *Flood Advisory* for Nueces County, the county of station KCRP). In the recent-alerts history one new id appeared and one week-old *Flash Flood Warning* was no longer returned: the endpoint is a rolling window, so absence does not mean deletion. The active-alerts `ETag` changed exactly when the content changed, so it can be used for conditional requests.

3.3 Revisions, deletions and consequences for the watermark

- **Observations** arrive about every 5 minutes, but only the hourly METAR reports carry `precipitationLastHou` values can arrive late or be corrected after quality control; therefore the watermark per station is the last `timestamp`, and every poll re-reads a 3-hour overlap and upserts by `(station, timestamp)`.
- **Alerts** are never edited in place: a change is a new message (new `id`) with `messageType` Update

or Cancel that references previous ids. Alerts also disappear from `/alerts/active` when they expire. Therefore the system keeps every message, chains them through `references` into one `event_group`, and marks disappeared alerts as inactive instead of deleting them.
- **Storm Events** (slow source) revises whole yearly files; the compile suffix is the watermark, and a changed suffix triggers the re-load of that year.

4 Source Profiling and Data Quality

The notebook `notebooks/01_source_profiling.ipynb` profiles at least 200 records per source (or the full response when smaller, which is the case of active alerts): null percentage per field, value ranges, duplicated identifiers, timestamp format and time zone, coordinate validity (Texas bounding box) and unit consistency.

4.1 Data-quality problems

The notebook found 11 concrete problems; each comes with a real example record:

Table 3: Data-quality problems found in the real samples (generated by the profiling notebook).

#	Source	Problem	Affected	Example record / design impact
1	`noaa_storm_events`	Local timestamps without offset; zone stored apart in CZ_TIMEZONE with several labels (CST-6, MST-7); two-digit year	724/724	*EVENT_ID 1256799: BEGIN_DATE_TIME='21-APR-25 10:30:00', CZ_TIMEZONE='CST-6'.* Convert to UTC using the offset in CZ_TIMEZONE before any comparison with other sources.
2	`noaa_storm_events`	Damage stored as text with K/M/B suffix; empty values mixed with '0.00K'	724/724	*EVENT_ID 1256799: DAMAGE_PROPERTY='0.00K' -> 0 USD.* Parse to numeric USD; keep empty as NULL (unknown), not 0.
3	`noaa_storm_events`	No physical severity measure for floods (MAGNITUDE empty); severity must be derived	724/724	*EVENT_ID 1256799 (Flood): MAGNITUDE empty.* Severity = rainfall from GHCN or impact (deaths/damage), not a native field.
4	`ncei_ghcnd`	Trace precipitation coded as PRCP=0 with measurement flag 'T'	232/1825	*USW00003927 2025-01-05: PRCP=0.0, attributes='T,,W'.* Keep the flag in the domain table; 0 mm with T is not the same as dry.
5	`ncei_ghcnd`	Daily value is a local-day total with no time zone; observation time in OBSTIME (values seen: (empty), 2400)	1825/1825	*USW00003927 2025-01-01: OBSTIME='(empty)', attributes=',,W'.* Match to events by local calendar day (+/- 1 day), not by exact UTC instant.
6	`nws_observations`	precipitationLastHour is null in many observations (value: null, unitCode wmoUnit:mm)	2463/2500	*KAUS 2026-11-06T02:55:00+00:00: precipitationLastHour=null, qc=nan.* Only the hourly (METAR) reports carry precipitation; sum precipitationLastHour from those, never from 5-min reports.

Table 3: Data-quality problems found in the real samples (generated by the profiling notebook).

#	Source	Problem	Affected	Example record / design impact
7	`nws_observations`	Mixed reporting intervals, not hourly (median 5 min; most common: 5 min x954, 15 min x205, 10 min x151)	2461/2500	*KAUS 2026-11-01T13:40:00+00:00: 5 min after previous observation.* Aggregate to fixed hourly/daily buckets; watermark per station on timestamp, not on count.
8	`nws_alerts`	Alerts without geometry (only zone/county codes in affectedZones, text in areaDesc)	89/226	*a938581ca63053af00abcf6f715cdb.001.1 Dense Fog Advisory: geometry=null, areaDesc='Caddo; Bossier; Webster; Claiborne; Lincoln; Union; De Soto;'.* Need NWS zone/county geometries to map alerts; join on UGC/SAME codes.
9	`nws_alerts`	Timestamps in local offsets, not UTC (offsets seen: -05:00, -06:00, -07:00)	226/226	*324c74ecf2274302ec477fdf42445c.001.1: sent='2026-11-05T20:40:00-06:00'.* Normalize every timestamp to UTC on ingestion.
10	`nws_alerts`	Same warning re-issued as Update/Cancel with a new id (chain via references)	111/226	*50cad2923b9d7c15f9df1b37f76079.001.1 Update Dense Fog Advisory: 1 reference(s).* Group alerts by reference chain; the latest message replaces earlier ones.
11	`nws_alerts`	Feed mixes all hazards; flood-related events must be filtered by 'event'	191/226	*7dae43678f154cccbf5d8510819d7d.001.1: event='Special Weather Statement'.* Filter on the event list in config/sources.yaml.

4.2 Overlap between sources

To verify that the sources describe the same real-world events, each Storm Events record with coordinates was matched to the nearest GHCN station within 60 km that recorded at least 25 mm on the event's local day or the previous day. 110 candidate pairs were found, covering 19 distinct station-days (several Storm Events records usually describe one storm, e.g. the San Antonio flash floods of 12 June 2025, when the Bexar County station recorded 155.2 mm). The strongest pair per station-day is:

Table 4: Candidate pairs: Storm Events record vs. GHCN-Daily station-day (nearest station within 60 km, ≥25 mm); strongest pair per station-day.

Storm EVENT_ID	Type	County	Begin (local)	Station / date	PRCP (mm)	Dist. (km)
1272305	Flash Flood	BEXAR	12-JUN-25 03:00:00	USW00012921 2025-06-12	155.2	0.7
1300295	Flash Flood	DALLAS	20-NOV-25 12:15:00	USW00003927 2025-11-20	100.3	12.5
1290922	Flash Flood	BEXAR	31-AUG-25 18:45:00	USW00012921 2025-08-31	79.0	1.4
1297755	Flash Flood	TARRANT	24-OCT-25 21:30:00	USW00003927 2025-10-24	78.7	32.0
1233051	Flash Flood	DALLAS	29-JAN-25 23:18:00	USW00003927 2025-01-29	71.1	6.4
1256260	Flash Flood	DALLAS	30-APR-25 11:00:00	USW00003927 2025-04-30	61.0	12.3
1264254	Flash Flood	TRAVIS	06-MAY-25 11:30:00	USW00013904 2025-05-05	55.1	27.5
1264618	Flash Flood	BEXAR	28-MAY-25 03:57:00	USW00012921 2025-05-28	52.1	4.3

What differs inside each pair: (1) *identifier*: there is no shared key (EVENT_ID vs. station-date); (2) *time*: an instant in local time with a separate zone label vs. a local-day total, so the match

needs a ±1 day window; (3) *position*: an event point (or only a county/zone) vs. a station point several km away; (4) *severity*: an impact description and damage in USD vs. physical rainfall in mm. For the incremental sources, flood alerts are linked to observation stations through the county names in `areaDesc` (Harris–KIAH, Travis–KAUS, Bexar–KSAT, Tarrant–KDFW, Nueces–KCRP).

5 Canonical Mapping

Rules: all timestamps in UTC; coordinates in WGS84 decimal degrees, NULL when the record is located only by county/zone (never 0,0); precipitation in mm; damage in USD (K = 10^3, M = 10^6, B = 10^9). Full table: `workshop_1/mapping.md`.

Table 5: Mapping from each source to the canonical `event`.

Canonical	Storm Events	GHCN-Daily	NWS observations	NWS alerts
event_id	`se:`EVENT_ID	`gh:`STATION:DATE	`obs:`stationId:timestamp	`al:`properties.id
source	noaa_storm_events	ncei_ghcnd	nws_observations	nws_alerts
event_type	EVENT_TYPE → flood / flash_flood / heavy_rain	daily_precipitation	precipitation_observation	event → flood_warning, flash_flood_watch, ...
timestamp	BEGIN_DATE_TIME + CZ_TIMEZONE offset → UTC	DATE (local day) → UTC, granularity = day	timestamp (UTC)	onset or effective → UTC
latitude / longitude	BEGIN_LAT / BEGIN_LON (NULL if zone)	LATITUDE / LONGITUDE	geometry.coordinates[1] / [0]	polygon centroid, else NULL
severity	derived: type + damage + casualties	PRCP (mm)	precipitationLastHour (mm)	event + severity
raw_reference	file name + EVENT_ID	request URL + STATION/DATE	@id URL	@id URL

Severity scale. `severity_level`: 0 none, 1 low, 2 medium, 3 high. Rainfall per day: <10, 10–25, 25–75, ≥75 mm; rainfall per hour: <2, 2–10, 10–25, ≥25 mm; alerts: Advisory/Statement = 1, Watch = 2, Warning = 3; Storm Events: Heavy Rain without damage = 1, Flood/Flash Flood with damage < 50K USD = 2, damage ≥ 50K USD or casualties = 3. These thresholds are preliminary and will be calibrated with the historical distribution.

Domain tables. `flood_event` (episode, county/zone code, flood cause, damage, casualties, narratives, source file and compile date); `daily_precipitation` (station, local date, PRCP mm, measurement/quality/source flags); `station_observation` (precipitation 1/3/6 h, QC code, temperature, description); `flood_alert` (message type, NWS event, severity, certainty, urgency, sent/expires/ends, UGC codes, references); and the static entity `weather_station` that pairs GHCN and NWS station identifiers. Figure 1 shows the first ER draft (DDL in `sql/schema_draft.sql`).

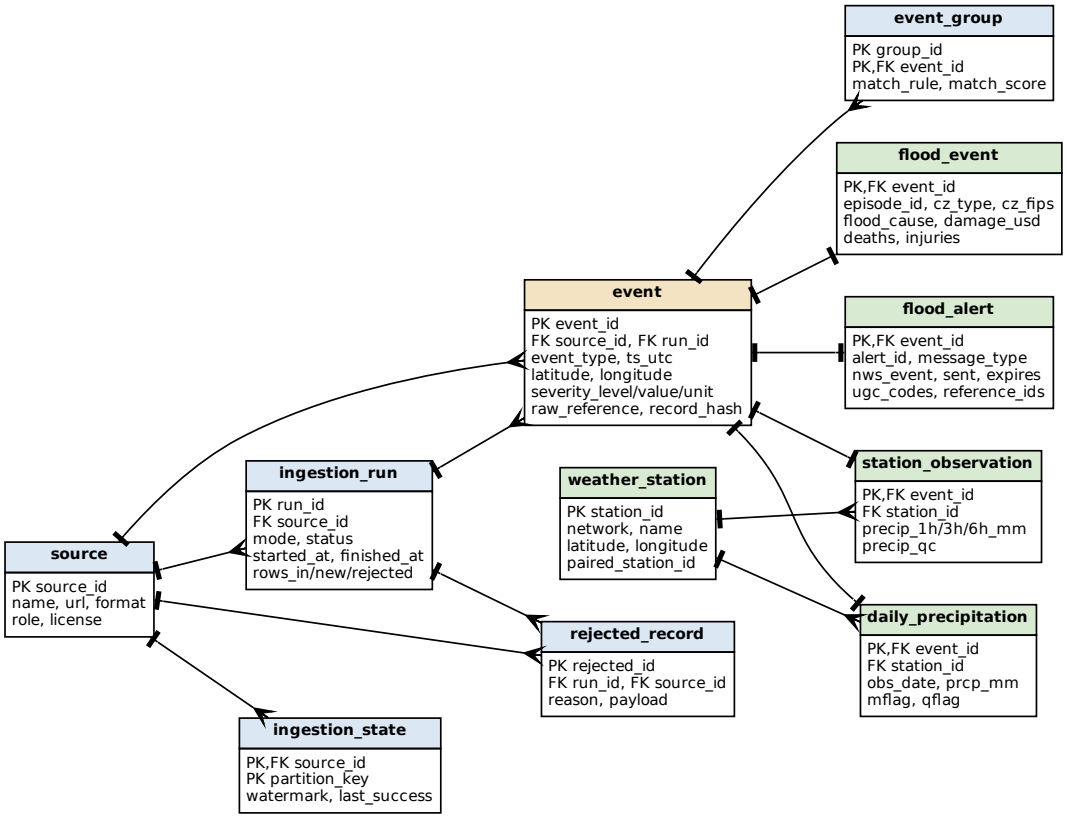


Figure 1: First draft ER diagram: common model (blue), canonical event (orange) and flood domain tables (green).

6 Ingestion Contract (Draft)

Table 6: Ingestion contract per source (details in `docs/ingestion_contract.md`).

Source	Historical strategy	Incremental strategy (watermark, frequency, overlap)	Duplicate key
Storm Events	Download each yearly file 2010–present once (cache), filter Texas + flood types; partition = year	Monthly listing check; watermark = `_cYYYYMMDD` per year; changed suffix ⇒ re-load that year; last 2 years always re-checked	EVENT_ID
GHCN-Daily	One request per station × year (85 requests)	Daily; watermark = max DATE per station; re-request last 30 days	(STATION, DATE)
NWS observations	Load the ∼7 days the API keeps	Every 60 min; watermark = max timestamp per station; request from watermark − 3 h	(stationId, timestamp)
NWS alerts	`/alerts?area=TX` with pagination for recent history	Every 15 min `/alerts/active?area=TX`; watermark = max `sent` + set of active ids; Update/Cancel chained by `references`; missing ids marked inactive	properties.id

Correctness when a run fails halfway. Each run (i) stores raw responses in the landing zone first, (ii) loads and validates them, and (iii) upserts each partition (year, station-year or poll) *in one transaction together with its watermark update*. A failure leaves finished partitions committed with their new watermark and the failed partition with the old one; the next run resumes from there. Because writes are upserts by the duplicate key, repeating a partition is idempotent. Invalid records go to `rejected_record` with the reason and payload instead of stopping the run, and the run is recorded as `failed` in `ingestion_run`.

7 Project Definition

7.1 Project overview

- **Title:** Flood Monitoring and Risk System for Texas.
- **Motivation:** flood information for Texas is spread across sources with different formats, time scales and geographic units (bulk CSV by county/zone, daily station totals, hourly observations, CAP alerts by forecast zone). Nobody can see in one place whether today's rain is unusual for a place with a history of flooding.
- **Goal (data engineering):** build a reproducible, idempotent pipeline that ingests two historical and two incremental sources into one canonical model, with watermarks, change detection, duplicate handling and quarantine. **Goal (analytics):** compare current rainfall and alerts with the historical rainfall distribution and flood frequency, and compute a Flood Risk Index per county.
- **Scope:** Texas; 2010–present; rainfall-driven flooding (Flood, Flash Flood, Heavy Rain); five paired stations, extendable through configuration.
- **Explicit exclusions:** no flood forecasting or hydrological modeling; no river-gauge (stage/discharge) data; no coastal storm-surge flooding; no real-time public warning service; no counties without

a nearby station in the first version.

- **Constraints:** free and open sources only; NWS requires a User-Agent; Storm Events delay of several months; Docker Compose on a student laptop; three team members; four workshops.

7.2 Functional user

Role: emergency-management analyst at a Texas county or regional office (e.g., Harris County). **Profile:** follows the weather daily, knows NWS products, is not a programmer, needs answers in minutes. **Decisions supported:** raise internal readiness, pre-position resources or close low-water crossings, prioritize which counties to watch, and brief officials with evidence. **Why the dashboard helps:** it puts current rainfall, its historical rarity, past floods in the same county and active alerts on one screen, and states how fresh and complete each source is.

7.3 User stories (MoSCoW)

ID	Story	Priority
US1	As an emergency analyst, I want to see the flood alerts active now in Texas on a map so that I know where official warnings are in effect.	Must
US2	As an emergency analyst, I want to see rainfall in the last 24 hours and 7 days per station so that I know where heavy rain is falling now.	Must
US3	As an emergency analyst, I want current rainfall compared with the historical percentile for the same station and month so that I can tell whether it is unusual.	Must
US4	As an emergency analyst, I want a Flood Risk Index (LOW/MEDIUM/HIGH) per county with the inputs that produced it so that I can prioritize counties and justify the decision.	Must
US5	As an emergency analyst, I want to see when each source was last updated and how many records were rejected so that I know how fresh and complete the data is.	Must
US6	As an emergency analyst, I want to see what is new since my last visit (new alerts, updates, cancellations) so that I do not miss changes.	Should
US7	As a risk researcher, I want the historical flood frequency and damage per county since 2010 so that I can identify recurrently affected areas.	Should
US8	As a risk researcher, I want to see which records from different sources were grouped as the same event so that I can trust the integration.	Could
US9	As an emergency analyst, I want an e-mail notification when a county turns HIGH.	Won't (this semester)

7.4 Indicator proposal: Flood Risk Index

Inputs (per county, at time t): R_{24} and R_{7d}, rainfall (mm) at the county's station; P_{24}, the percentile of R_{24} in the station's historical distribution for the same month (GHCN-Daily 2010–present); A, the highest active NWS flood alert level (0 none, 1 advisory/statement, 2 watch, 3 warning); F, the historical flood frequency of the county (Storm Events flood records per year, 2010–present).

Rule (preliminary):

$$\text{rain_level} = \begin{cases} 3 & P_{24} \geq 99 \text{ or } R_{24} \geq 75 \text{ mm} \\ 2 & P_{24} \geq 95 \text{ or } R_{7d} \geq 100 \text{ mm} \\ 1 & P_{24} \geq 90 \\ 0 & \text{otherwise} \end{cases}$$

$$\text{score} = \max(A, \text{rain_level}) + \mathbb{1}[F \text{ in the top quartile of Texas counties}]$$

Output scale: LOW (score 0–1), MEDIUM (2), HIGH (≥3), shown together with its inputs and the freshness of each source. The index is a *monitoring* aid, not a prediction of flooding.

Validation against history: alerts are not available for past years, so the rainfall + frequency part is back-tested with GHCN-Daily and Storm Events for 2010–present: for every county-day we compute the index and check (i) the share of Storm Events flood days classified MEDIUM/HIGH (hit rate) and (ii) the share of MEDIUM/HIGH days without any flood record (false-alarm rate). Thresholds are tuned on 2010–2020 and tested on 2021–present.

7.5 Risks and mitigation

Risk	Impact	Mitigation
Source outage (NWS API down, NCEI maintenance)	Incremental runs fail; dashboard shows stale data	Retries with backoff; run marked failed and resumed from the watermark; freshness shown on the dashboard; recorded responses in `tests/fixtures` for development
Rate limits / blocked client	HTTP 403/429, missing data	Descriptive User-Agent, polling no faster than 15 min, cache all downloads, never re-download the archive
Schema change (new columns, API version, renamed file pattern)	Parsing errors	Schema validation before loading; unknown or invalid records to `rejected_record`; source-status log and notice to the professor
Late revisions (Storm Events re-compiled, observations corrected)	Silent inconsistencies	Compile-suffix watermark per year; overlap windows; content hash per record
Ambiguous identifiers and locations (zone vs. county, alert chains, no shared key)	Wrong or missing event groups	UGC/FIPS lookup tables; matching rules with distance and time thresholds stored in `event_group.match_rule`; manual review of samples
Data volume / laptop resources	Slow loads, full disk	Texas and flood-type filtering at load; partitioning by year and station; raw files git-ignored

Risk	Impact	Mitigation
License / attribution	Non-compliance	All sources public domain; attribution recorded in `sources.yaml` and the report

7.6 Architecture overview

Figure 2 adapts the reference architecture of the course project (Sierra, 2026): sources → raw landing zone → normalize/validate/deduplicate → PostgreSQL/PostGIS (with quarantine) → analytics and indicator → Plotly Dash. Everything runs with Docker Compose; Apache Airflow schedules the incremental polls.

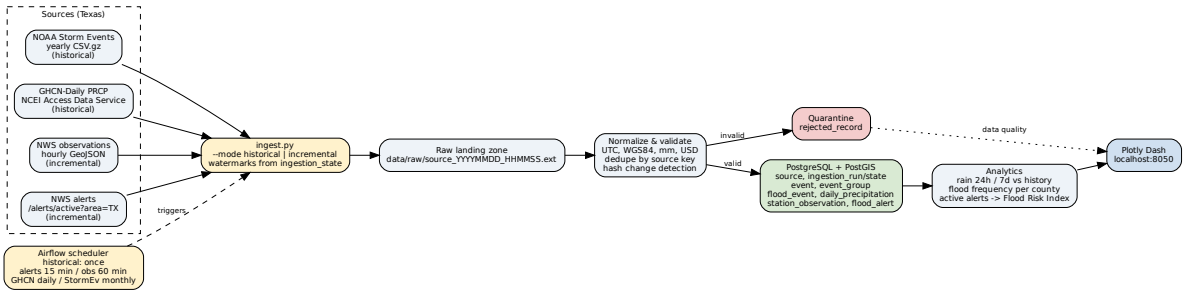


Figure 2: High-level architecture of the Flood Monitoring and Risk system.

8 Repository

Repository: `https://github.com/natcastill/flood-monitoring-team2`. Structure: `src/flood_monitor/` (config, HTTP client, one module per source), `ingest.py` (CLI stub: `-mode historical | incremental | status`), `config/sources.yaml`, `data/raw/` (git-ignored), `sql/schema_draft.sql`, `dashboard/`, `tests/` and `tests/fixtures/`, `notebooks/01_source_profiling.ipynb`, `docs/` (ingestion contract, ER and architecture diagrams, source-status log), `workshop_1/` (`mapping.md`, `samples/`), `.env.example`, `.gitignore` and `README.md`. No secrets are committed: the NWS contact is read from `.env`. The large Storm Events yearly files are not committed; `scripts/fetch_samples.py` downloads them into `data/raw/`.

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